Xingyu Wang

Contact Information

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Research Interests

Rare Event Analysis, Large Deviations, Metastability, Heavy Tails; Machine Learning; Stochastic Simulation; Self-exciting Processes

Academic Positions

Antai College of Economics and Management, Shanghai Jiao Tong University — Shanghai, China
Assistant Professor — 2026 - Present
Department of Management Science

University of Amsterdam — Amsterdam, Netherlands
Postdoc Researcher — 2024 - 2026
Quantitative Economics, Amsterdam School of Economics
Hosts: Roger J.A. Laeven, Bert Zwart

Education

Northwestern University — Evanston, IL, United States
Ph.D., Department of Industrial Engineering and Management Sciences — 2018 - 2024
Advisor: Chang-Han Rhee

Northwestern University — Evanston, IL, United States
M.S. in Analytics — 2016 - 2017

Peking University — Beijing, China
B.S. in Psychology and Applied Mathematics — 2011 - 2016

Journal Papers
($\alpha - \beta$: alphabetical authorship)

[1] Strongly Efficient Rare-Event Simulation for Regularly Varying Lévy Processes with Infinite Activities
Xingyu Wang, Chang-Han Rhee
Mathematics of Operations Research, Accepted

[2] Sample Path Large Deviations for Multivariate Heavy-Tailed Hawkes Processes and Related Lévy Processes
($\alpha - \beta$) Jose Blanchet, Roger J. A. Laeven, Xingyu Wang, Bert Zwart
Bernoulli, Accepted

[3] Locally Uniform Sample-Path Large Deviations for Truncated Heavy-Tailed Dynamical System
Xingyu Wang, Chang-Han Rhee
Annals of Applied Probability, Accepted
George Nicholson Student Paper Competition, Second Place, 2023

Submitted and Working Papers

[4] Global Dynamics of Heavy-Tailed SGDs in Nonconvex Loss Landscape: Characterization and Control
Xingyu Wang, Chang-Han Rhee
Major Revision at *Operations Research*; arXiv:2510.20905

[5] Tail Asymptotics of Cluster Sizes in Multivariate Heavy-Tailed Hawkes Processes
($\alpha - \beta$) Jose Blanchet, Roger J. A. Laeven, Xingyu Wang, Bert Zwart
Major Revision at *Annals of Applied Probability*; `arXiv:2503.01004`

Conference Papers

[6] Multi-agent Multi-armed Bandit with Fully Heavy-tailed Dynamics
Xingyu Wang, Mengfan Xu
Winter Simulation Conference (WSC), 2025

[7] Importance Sampling Strategy for Heavy-Tailed Systems with Catastrophe Principle
Xingyu Wang, Chang-Han Rhee
Winter Simulation Conference (WSC), 2023

[8] Eliminating Sharp Minima from SGD with Truncated Heavy-Tailed Noises
Xingyu Wang, Sewoong Oh, Chang-Han Rhee
International Conference on Learning Representations (ICLR), 2022
***Nemhauser Prize for Best Student Paper**, 2022*

[9] Keyword-Based Topic Modeling and Keyword Selection
Xingyu Wang, Lida Zhang, Diego Klabjan
IEEE Big Data, 2021

[10] Efficient Rare-Event Simulation for Multiple Jump Events in Regularly Varying Lévy Processes with Infinite Activities
Xingyu Wang, Chang-Han Rhee
Winter Simulation Conference (WSC), 2020

[11] Competitive Multi-Agent Inverse Reinforcement Learning with Sub-Optimal Demonstrations
Xingyu Wang, Diego Klabjan
International Conference on Machine Learning (ICML), 2018

Honors and Awards

- **Danish Data Science Academy Visit Grant**, 2026
- **Nemhauser Best Dissertation Prize**, Northwestern University 2025
- **George Nicholson Student Paper Competition, Second Place**, INFORMS 2023
- **Terminal Year Fellowship**, Northwestern University 2023
- **Nemhauser Prize for Best Student Paper**, Department of Industrial Engineering and Management Sciences, Northwestern University 2022
- **Benjamin A. Sachs Graduate Fellowship**, Northwestern University 2022
- **Arthur P. Hurter Award for Academic Excellence among First Year Graduate Students**, Department of Industrial Engineering and Management Sciences, Northwestern University 2019
- **Lee Wai Wang Scholarship**, Department of Psychology, Peking University 2015
- **National Scholarship for Undergraduates**, Department of Psychology, Peking University 2013

Talks

Large Deviations for Multivariate Heavy-Tailed Hawkes Processes

- IISE Annual Conference & Expo, Arlington, TX May, 2026
- INFORMS Annual Meeting, Atlanta, GA Oct, 2025

- Seminar at School of Mathematics and Statistics, Beijing Institute of Technology — Apr, 2025
- Stochastic Seminars at Korteweg-de Vries Institute for Mathematics, Amsterdam, Netherlands — Feb, 2025

Sharp Characterization and Control of Global Dynamics of SGDs with Heavy Tails

- University of Copenhagen, Statistics Seminar — Sep, 2025
- Monte Carlo Methods 2025, Chicago, IL — July, 2025
- Applied Probability Society Conference 2025, Atlanta, GA — July, 2025
- Bayes Comp 2025, Singapore — Jun, 2025
- Data-Driven Queueing Challenges Conference, Eindhoven, Netherlands — Nov, 2024
- INI Satellite Programme on Heavy Tails in Machine Learning, London, United Kingdom — Apr, 2024
- Cornell University, ORIE, Ithaca, NY — Feb, 2024
- University of Pittsburgh, Industrial Engineering, Pittsburgh, PA — Jan, 2024
- SNAPP Seminar, Lightning Talk Session, Virtual — Dec, 2023

Importance Sampling Strategy for Heavy-Tailed Systems with Catastrophe Principle

- Winter Simulation Conference (Advanced Tutorial), San Antonio, TX — Dec, 2023

Large Deviation and Metastability Analysis for Heavy-Tailed Dynamical Systems

- Neurips 2023, Heavy Tails in ML Workshop (Poster), New Orleans, LA — Dec, 2023
- INFORMS Annual Meeting, Phoenix, AZ — Oct, 2023
- Cornell ORIE Young Researchers Workshop, Ithaca, NY — Oct, 2023

Eliminating Sharp Minima from SGD with Truncated Heavy-Tailed Noises

- INFORMS Annual Meeting, Phoenix, AZ — Oct, 2023
- Applied Probability Society Conference 2023, Nancy, France — Jun, 2023
- INFORMS Annual Meeting, Indianapolis, IN — Oct, 2022
- International Conference on Learning Representations (2022), Virtual — Apr, 2022
- DeepMath (2021), Virtual — Nov, 2021

Efficient Rare-Event Simulation for Multiple Jump Events in Regularly Varying Lévy Processes with Infinite Activities

- Winter Simulation Conference, Virtual — Dec, 2020
- INFORMS Annual Meeting, Virtual — Nov, 2020

Teaching

Theses Supervisor

- UvA, Master's Theses, Actuarial Science and Mathematical Finance — 2025, 2026
- UvA, Bachelor's Theses, Actuarial Science — 2025, 2026

Instructor

- IEMS Bootcamp (Probability) for PhD Students 2022 Fall, 2023 Fall

Teaching Assistant

- UvA Introduction to Data Science 2025
- UvA Non-Life Insurance: Statistical Techniques and Data Analytics 2024, 2025
- NU IEMS 317 Discrete Event Systems Simulation 2022 Winter; 2021 Winter
- NU IEMS 315 Stochastic Models 2021 Spring, Fall; 2020 Spring, Fall

Course Grader

- NU OPNS 450 Decision Models & Prescriptive Analytics 2022 Winter
- NU IEMS 435 Stochastic Simulation 2020 Winter

Services

- Referee: *Operations Research, Management Science, Mathematics of Operations Research, INFORMS Journal on Computing, Journal of Machine Learning Research, Communications Physics, Computers and Operations Research, Journal of Simulation, ICML (2025), Winter Simulation Conference (2023, 2025)*
- Volunteer:
  - *NU MORE-REACH panel (2024)*
- Webmaster:
  - *Stochastic Networks, Applied Probability, and Performance (SNAPP) seminar (2023 - 2024)*
- Mentor:
  - *1st-year mentorship program at NU INFORMS Student Chapter (2020, 2021)*

Last Update: October 25, 2026